Linear graphs

GRAPHS ARE A WAY OF PICTURING AN EQUATION. A LINEAR EQUATION ALWAYS HAS A STRAIGHT LINE.

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Graphs of linear equations

A linear equation is an equation that does not contain a squared variable such as x^2, or a variable of a higher power, such as x^3. Linear equations can be represented by straight line graphs, where the line passes through coordinates that satisfy the equation. For example, one of the sets of coordinates for $y = x + 5$ is (1, 6), because $6 = 1 + 5$.

slope, or gradient

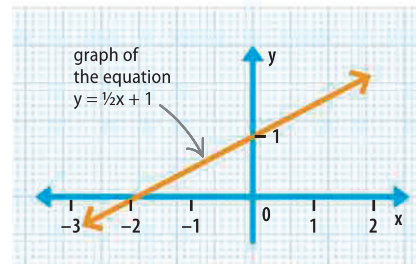
value of x

$$y = mx + b$$

value of y

value of y intercept—the point where the line crosses y axis

△ **The equation of a straight line**
All straight lines have an equation. The value of m is the slope (or slope) of the line and b is where it cuts the y axis.



△ **A linear graph**
The graph of an equation is a set of points with coordinates that satisfy the equation.

Finding the equation of a line

To find the equation of a given line, use the graph to find its slope and y intercept. Then substitute them into the equation for a line, $y = mx + b$.

To find the slope of the line (m), draw lines out from a section of the line as shown. Then divide the vertical distance by horizontal distance—the result is the slope.

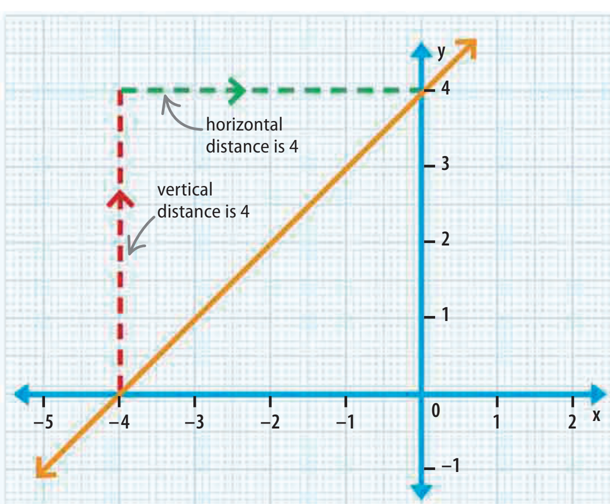
$$\text{slope} = \frac{\text{vertical distance}}{\text{horizontal distance}} \rightarrow \frac{4}{4} = +1$$

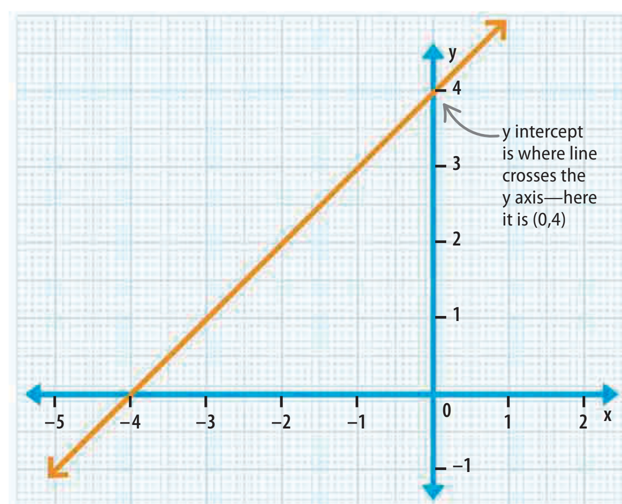
slope

division sign

To find the y intercept, look at the graph and find where the line crosses the y axis. This is the y intercept, and is b in the equation.

$$y \text{ intercept} = \mathbf{(0,4)}$$





Finally, substitute the values that have been found from the graph into the equation for a line. This gives the equation for the line shown above.

slope is +1

y intercept is 4

$$y = \underbrace{mx}_{\text{slope is } +1} + \underbrace{b}_{\text{y intercept is 4}} \Rightarrow y = x + 4$$

1x simplifies to x